A Learning Style-aware Dropout Prediction Method via Fusing Global and Local Semantics

Guiqiang Luo[1,2,3], Weizhong Zhao[1,2,3*], Xingpeng Jiang[1,2,3], Tingting He[1,2,3]
[1]*Hubei Provincial Key Laboratory of Artificial Intelligence and Smart Learning, Central China Normal University, Wuhan 430079, Hubei China*
[2]*School of Computer, Central China Normal University, Wuhan 430079, Hubei China*
[3]*National Language Resources Monitoring & Research Center for Network Media, Central China Normal University, Wuhan 430079, Hubei China*

2024 International Conference on Intelligent Education and Intelligent Research (IEIR) | 979-8-3315-1982-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/IEIR62538.2024.10960124

***Abstract*—Massive Open Online Courses (MOOCs), with their vast resources and significant flexibility, have emerged as the most popular way for online education globally. However, one of the emergent challenges faced by numerous MOOC platforms is the high rate of dropout. While many approaches have been proposed to predict dropout rates, the common drawback is that they usually ignore the theories from educational psychology when modeling the behavior from log data, deriving undesirable performance for the task of dropout prediction. In this paper, we propose a novel learning style-aware framework for predicting dropout on MOOCs, effectively fusing global and local semantics contained in learners' learning behaviors. First, the learning styles are identified by clustering on time series data in logs of MOOCs. Then, the global and local semantics contained in the learning style information and learning behavior information are captured by a module of CNNs, deriving the meaningful hidden representations. Finally, the obtained high-quality representations are used to derive the predicted dropout behavior given learner-course pairs. Extensive experiments are carried out on the KDD Cup 2015 dataset, and the results demonstrate the effectiveness of the proposed method.**

***Keywords*—Dropout prediction, Learning style detection, Dynamic time warping.**

I. Introduction

Since its advent in 2006, Massive Open Online Courses (MOOCs) have experienced rapid and robust growth [1]. Public online educational platforms, such as Coursera (www.coursera.org) and edX (www.edx.org), have aggregated a rich collection of high-quality, free courses from globally renowned universities. Moreover, those platforms are able to conveniently provide learners with online support, which includes task assignments, learning assessments, and interactive communication among teachers and learners [2]. Comparing with the traditional pattern of education in classroom, the main strengths of MOOCs lie in the following three aspects [3]. 1). It enables learners with different backgrounds to make their own plans for learning courses, enhancing the efficiency of online learning; 2). Since it breaks the limits of time and space, online learners can access multiple materials or resources actively, guaranteeing the effectiveness of online learning; 3). Owing to the development of information technologies, online courses are accessible for a diverse range of devices, ensuring the high scalability of educational resources. Therefore, MOOCs have been becoming popular around the whole world, and revolutionizing the field of education as well [4].

Despite the continuous surge in registration numbers in MOOCs, however, the proportion of learners who can successfully complete their registered courses remains extremely low. Statistics show that among those enrolled in MOOCs, more than 90% of students fail to complete the courses, and the retention rate only falls in the range of 3% to 15% [5]. The dropout behavior is detrimental to both educational institutions and learners [6]. For educational institutions, a high dropout rate not only results in the waste of educational resources but also impacts their reputation in users, subsequently affecting their financial interests. For learners, the occurrence of dropout not only leads to the waste of time but also might have adverse effects on learners' psychological aspects such as self-confidence, perseverance, and determination. To address this issue of high dropout rate, a feasible solution is to accurately predict the dropout behavior of learners in advance in the study process of registered courses [7]. Once monitoring the signal of dropout, appropriate intervention(s) can be provided in time to help students catch up with normal progress of courses, and successfully complete their studies. Consequently, predicting dropout on MOOCs has become a hot research topic in the field of intelligent education or smart education in recent years.

For MOOCs, their log data typically record the sequences of actions left on the platforms by learners when studying registered courses [8]. More specifically, the clickstream data of log record users' behavior in the process of studying courses, including watching videos, completing assignments, participating in forum discussions, etc. Based on the log data, researchers have proposed many different methods for predicting dropout on MOOCs according to diverse strategies [9]. Existing methods can be roughly classified into two categories, traditional machine learning-based methods and deep learning-based methods. For traditional machine learning-based methods, significant features are first extracted from log data through a process called feature engineering. Then, based on the obtained features, classical classification models are trained, such as logistic regression (LR) [10], and support

*Correspondence: Dr. Weizhong Zhao, Email: wzzhao@ccnu.edu.cn

979-8-3315-1982-7/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

vector machine (SVM) [11]. Finally, the possible dropout behavior is predicted given each pair of a learner and a registered course. For deep learning-based methods, deep neural networks are used to learn hidden features from log data in an end-to-end manner. The representative deep learning-based methods include the CNN-based models [12] and LSTM-based models [13].

Although some of existing methods derive relatively acceptable performance on the task of dropout prediction, the common drawback is that knowledge from educational psychology is not explicitly employed when modeling the behavior from log data, leaving room to enhance the prediction performance [14]. According to theories in educational psychology, each learner might possess distinct learning styles, which will be reflected in their behavior during the learning process [15]. This is also a typical phenomenon in the online MOOCs platforms. For instance, some learners like to distribute their studying hours every day, while some learners prefer to conduct more concentrated studies over the weekends [16]. Similarly, for advancing the learning effect, some learners like to join the discussion forum for courses, while some learners prefer to watch course videos in multiple times. Therefore, considering specific learning styles among learners will derive more meaningful features for students' learning behaviors, improving the performance of dropout prediction accordingly.

Based on the observations mentioned above, in this paper, we propose a novel framework for predicting dropout on MOOCs, in which students' learning styles are considered explicitly in behavior modeling. More specifically, the learning styles are first obtained by clustering time series data in log of MOOCs, by which the global semantics contained in the whole time span of learning process can be captured elegantly. Then, a module of CNNs is employed on the concatenation of learning style information and various types of learning behavior information to derive the meaningful hidden representations, effectively fusing global and local semantics contained in the log data. Finally, the predicted dropout behavior can be derived based on the high-quality representations of given learner-course pairs. Generally, the main contributions of this paper are summarized as follows.

- Learners' learning styles are detected by performing a time series clustering algorithm on the sequential log data, capturing the global semantics contained in learners' behavior of the whole learning process.
- A learning style-aware dropout prediction model is proposed, in which the global and local semantics are effectively used by a module of CNNs, and the advanced prediction performance is obtained accordingly.
- Extensive experiments are carried out on a widely used real-world dataset, and the results demonstrate the effectiveness of the proposed method for dropout prediction.

II. Related work

In this section, we briefly review the related work to our study, including the applied definition of dropout, existing dropout prediction methods, and methods of learning style detection.

TABLE I: Toy example for illustrating the difference of three definitions of dropout.

Time	W1	W2	W3	W4	W5
Log	[5,23,6,5]	None	None	[4,2,6,10]	None
Def. 1	1	1	1	1	1
Def. 2	0	0	0	1	1
Def. 3	1	1	0	1	N.A.

A. Definition of Dropout

In the field of intelligent education, there are three applied definitions of dropout. The difference of three definition is illustrated in Table I. For the toy example in Table I, the course spans in five weeks. For each week, the log records the frequencies of four types of behaviors. None denotes there is no behavior in the corresponding time span. In addition, "1" denotes the label "dropout", and "0" denotes the label "non-dropout". The first definition assigns labels for each learner-course pair according to if the learner engages in the course (i.e., leaving interaction record) on MOOCs or not during the last week (i.e., W5 in Table I) of the course [17]–[19]. If there is no behavior in the last week, then the labels in all weeks are assigned with "dropout"; otherwise, the labels in all weeks are assigned with "non-dropout". The second definition assigns labels according to the time when the learner has the last interaction with the MOOCs [11], [20], [21]. All the weeks (e.g., the first three weeks in Table I) before the last interaction are labeled as "non-dropout", while the remaining weeks (e.g., the 4-th week and the 5-th week in Table I) are labeled as "dropout". The third definition assigns the label for the current week based on the log records in the next week [22]. If there is no activity in the next week, then the label of the current week is "dropout"; otherwise, the label of the current week is "non-dropout". It is simple to find that for the third definition, there is no label for the last week (i.e., W5 as illustrated in Table I).

Since the third definition of dropout is more accurately to describe the learning status of students in every week and is widely used in the field of intelligent education, we adopt the third definition as well in this study. In the following subsection, we review briefly existing work on dropout prediction in the line of the third definition.

B. Existing Dropout Methods

For traditional machine learning-based methods, the representative dropout models include FSPred [10], WSVM [11], SPRF [23]. For FSPred, different types of learning behavior are extracted as features, on which a LR is trained as classifier for predicting dropout [10]. For WSVM, features of activity levels in log data are extracted to train a SVM, by which the dropout behavior is predicted for each learner-course pair [11]. In SPRF, features of students' daily learning progress are extracted, and

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

then a Random Forest (RF) is trained as a classifier, by which the predicted probabiltiy of dropout behavior can be derived for each learner-course pair [23].

For deep learning-based methods, the representative dropout models include NSSM [13] and LCNN [12]. Inspired by LSTM, in NSSM, hidden state is modeled for each time step from students' learning logs, and dropout behavior is predicted based on the representations of current and previous states [13]. In LCNN, a novel Convolutional Neural Network (CNN) model is used to capture the local semantics contained in students' behavior from the log data, and the dropout behavior is predicted on the obtained features accordingly.

The existing methods employ directly the historical behavior log data to learn or extract features for predicting dropout behavior, and fail to utilize the educational psychology theories such as users' learning styles, deriving sub-optimal prediction performance.

C. Methods of Learning Style Detection

Learning style detection refers to the procedure of using machine learning algorithms or educational technologies to identify and understand the preferred learning styles of individual learners within the MOOC platforms [24]. The representative methods are presented as follows.

Muhammad et al. (2022) [24] proposed a learning style detection method based on graph representation learning, in which a bipartite graph is first constructed to describe the interactions between the set of learners and the set of learning resources, and then graph embedding is learned for learners and courses, finally the learning styles are derived based on the representations of learners. Zhang et al. (2020) [25] utilized the deep belief network to learn hidden representations for students' learning behavior, and then the learning styles can be predicted accordingly. Hidayat et al. (2020) [26] proposed a learning style detection model, in which 29 features are first extracted from log data, and then the learning styles of students are detected by combining a modified K-Means algorithm with the Naive Bayes classification algorithm.

According to above analysis, we propose a learning style-aware deep learning-based model for dropout prediction, in which time series clustering algorithm is employed to obtain learning styles from behavior log data, and a module of CNNs is used to learn meaningful representations for learners and courses. With the high-quality representations, the performance of dropout prediction is enhanced accordingly. The details of the proposed method are presented in the following section.

III. The Proposed Method

As illustrated in Figure 1, our proposed method comprises four modules, i.e., preprocessing module, learning style detection module, behavior modeling module, and dropout prediction module. In preprocessing module, for each pair of learner and course, the frequencies of different behavior types in each week are computed from logs, and the form of time series data is obtained. Then, students' learning styles are identified by the learning style detection module. Next, in behavior modeling module, the hidden features of learners' behavior are obtained based on the combination of learners' time series data and learning styles. Finally, the dropout prediction is conducted on the derived features for given pairs of learner and course. Each module will be described as follows.

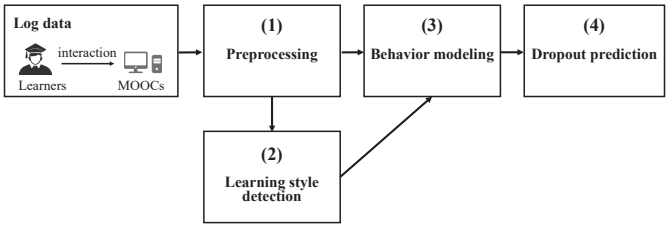


Fig. 1: The Schematic Flowchart of the Proposed Model.

A. Preprocessing

It is known that each learner on MOOCs can register several courses, and each course can be registered or selected by many learners. Moreover, each course usually lasts several weeks. For MOOCs platforms, the log file records various types of interactions between learners and platforms, including watching videos, completing assignments, engaging in forum discussions, and etc. Therefore, in order to predict the dropout behavior of learners, behavior records in log are required to be extracted and transformed into the manner of time series data.

Given each registration (i.e., a pair of learner and the selected course), we compute the frequencies of different types of behaviors for each week. Then, for each week, the learning record is represented by a vector, each component of which denotes the frequency of the corresponding behavior type. Assume the total number of behavior types in a MOOCs platform is denoted by N^b, and the maximum number of weeks of courses is denoted by T. For the registration of learner i on course j, the behavior record is formally denoted by following feature matrix.

$$s_{i,j} = \begin{pmatrix} s_{i,j,1}^{(1)} & \cdots & s_{i,j,1}^{(t)} & \cdots & s_{i,j,1}^{(T)} \\ s_{i,j,2}^{(1)} & \cdots & s_{i,j,2}^{(t)} & \cdots & s_{i,j,2}^{(T)} \\ \vdots & \vdots & \ddots & \vdots & \\ s_{i,j,N^b}^{(1)} & \cdots & s_{i,j,N^b}^{(t)} & \cdots & s_{i,j,N^b}^{(T)} \end{pmatrix} \tag{1}$$

It is clear that the t-th column of the feature matrix $s_{i,j}$ is frequencies of different behavior types in the t-th week.

According to the third definition provided in subsection II-A and the log file, the dropout label for the registration of learner i on course j can be obtained and denoted as follows.

$$y_{i,j} = \left(y_{i,j}^{(1)}, \ldots, y_{i,j}^{(t)}, \ldots, y_{i,j}^{(T)}\right) \tag{2}$$

Here, $y_{i,j}^{(t)} \in \{0, 1\}$. If the learner i drops out the course after the t-th week when studying course j, $y_{i,j}^{(t)} = 1$; otherwise, $y_{i,j}^{(t)} = 0$.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

B. Learning Style Detection

In this study, the basic idea for learners' learning style detection is to classify learners based on the time series data of their learning behaviors on MOOCs platforms. Based on the theories of educational psychology, students' learning styles will be reflected in the interactions between students and the online platforms. Therefore, clustering learners based on the behavior features can obtain subgroups of students, each of which exhibits the similar learning styles for the students in the corresponding subgroup.

For the module of learning style detection, it consists of two steps, i.e., obtaining the (hidden) features of students, and clustering students based on the obtained features. More specifically, a Dynamic Time Warping (DTW) based K-means clustering method is utilized to derive a collection of K learner subgroups, which is denoted by $C = \{C_1, \cdots, C_k, \cdots, C_K\}$.

1) Derivation of Learners' Features: In this study, we employ two methods to derive the features for learning style detection. For the first method, the original time series (OTS) data are used to derive features of students. Due to the possible issue of sparseness for the OTS-based method, in the second method, the graph autoencoder (GAE) [27] is utilized to derive hidden features of students.

OTS-based Method

Given the time series feature matrix derived from the preprocessing module, in the OTS-based method, the features of each learner are obtained by the average of feature matrices for all registrations made by the learner. Formally, for each learner $i \in \{1, \cdots, N_l\}$, the features S_i is defined as follows.

$$S_i = \frac{1}{n_i} \sum_{j=1}^{n_i} s_{i,j} \tag{3}$$

in which n_i denotes the total number of registered courses by the student i, and $s_{i,j}$ represents the original feature matrix of learner i on the j-th course (as shown in Eq. 1).

Because the obtained OTS-based features might be sparse, which poses a great challenge for clustering methods, we introduce the GAE-based method to derive dense representations of students, which is presented as follows.

GAE-based Method

Generally, the GAE-based method for deriving learners' features consists of three steps, i.e., L-C Bipartite Graph Construction, Graph Encoder, and Graph Decoder. In the first module, we construct a L-C bipartite graph that characterizes the interactions between learners and courses, and the initial features for both types of nodes are calculated as well. The graph encoder involves transforming the initial representations of learners and courses into hidden representations, which will be used for detecting learners' styles. The graph decoder is used to reconstruct the L-C bipartite graph for training the graph autoencoder model.

L-C Bipartite Graph Construction In our study, the constructed learner-course bipartite graph is denoted by $G = \{L, C, A\}$, which describes the interactions between learners and courses. Specifically, L denotes the set of N_l learners, C denotes the set of N_c courses, and $A \in \mathbb{R}^{(N_l+N_c)\times(N_l+N_c)}$ denotes the adjacency of the bipartite graph. Note that A is a weighted symmetric matrix with zero diagonal elements, illustrating the absence of connections between learners and learners, and between courses and courses; the weighted connections between learners and courses in A are computed based on the total number of interactions between the corresponding learner-course pair. In addition, the initial features of nodes are denoted by $X \in \mathbb{R}^{(N_l+N_c)\times N_b}$. Specifically, the initial features of learners are computed by the average frequency of each type of learning behavior associated with all registrations made by the learner, and the initial features of courses are represented by the average frequency of each type of learning behavior in all users' registrations of the course.

The adjacency matrix A and the initial features X will be input into the graph encoder to derive hidden representations of learners and courses.

Graph Encoder Given the input (A, X), the graph encoder uses a two-layer graph convolution networks (GCN) to derive the hidden representations of graph nodes. Formally, the GAE encoder is defined as follows.

$$GAE(A, X) = ReLU(\tilde{A} \cdot ReLU(\tilde{A} \cdot X \cdot W_0) \cdot W_1) \tag{4}$$

in which $\tilde{A} = D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$, and W_0 and W_1 are the two trainable parameters of the GCN layers.

The latent representations derived by graph encoder are denoted as Z, i.e., $Z = GAE(A, X)$ where $Z = \mathbb{R}^{(N_l+N_c)\times f}$, and f represents the dimension of embedding space. It is simple to know that the first N_l rows of Z are extracted to derive the hidden representations of N_l learners. To be consistent with denotations of OTS-based method, the hidden representations of learners are denoted by $\{S_1, \ldots, S_i, \ldots, S_{N_l}\}$.

Graph Decoder In graph decoder, the operation of dot product is conducted on the hidden representations Z to reconstruct the adjacency matrix, which is denoted by A'.

$$A' = \sigma(Z \cdot Z^\top) \tag{5}$$

To train the graph encoder, the cross-entropy loss is employed to evaluate the difference between the reconstructed adjacency matrix A' and the initial adjacency matrix A. Formally, the loss is defined as follows.

$$\mathcal{L}_{\text{GAE}} = -\frac{1}{N_l \cdot N_c} \sum_{i,j} (A_{i,j} \log A'_{i,j} + (1 - A_{i,j}) \log(1 - A'_{i,j})) \tag{6}$$

It is worth noting that the GAE can be trained in an offline manner. It means the learners' features can be obtained in advance by applying the GAE-based method.

2) Dynamic Time Warping-based K-means Clustering: Since the learning behavior data is in the pattern of time series, an effective learning style detection method requires to consider the semantics in the sequential data. In this study, a version of K-means on time series data is employed to capture the learning styles of learners, in which the metric of Dynamic Time Warping (DTW) [28] is utilized to determine the distance between two samples in the pattern of time series.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

Given two sequence samples (e.g., each sample is a learner's records of N_w weeks for different types of behavior), the essence of DTW is to find an optimal match between two given sequences. These two sequences are "warped" non-linearly in the time dimension to determine a measure of their similarity, irrespective of certain variations in time. This warping procedure allows DTW to be robust to differences in temporal alignment. Given representations of two learners which are denoted by S_i and S_j, the distance according to DTW is denoted by $D^{dtw}(S_i, S_j)$.

In this study, a DTW-version of K-means method is conducted on the learners' features obtained from the subsection III-B1. The challenge is how to update the mean or centroid of each cluster in which the time dimension is considered. The basic idea is to find a representative sequence or sample for each cluster that minimizes the total DTW distance between itself and all other sequences in the cluster. This representative sequence acts as the "mean" or centroid of the cluster. In our implementation, the DTW barycenter averaging (DBA) [28] approach is used for updating the means of clusters.

Note that the elbow method [29] is used to determine the optimal number of clusters in our experiments. In addition, given the learners' features obtained by OTS-based method, the sequence length T is the max number of lasting weeks for courses. However, for the learners' features obtained by GAE-based method, the sequence length T is 1, and the learning style detection method degenerates to the standard K-means. It means that the learning style detection method ignores the semantics in the sequential context, and the experimental results in the next section demonstrate that this drawback will affect the performance on dropout prediction.

C. Behavior Modeling via CNNs

After the previous step, each learner is assigned with a cluster ID which represents the specific learning style of the learner. In order to process the learning style information conveniently, the corresponding cluster ID is denoted by the one-hot embedding, i.e., a K-dimensional vector with only one component is 1 and other components are all 0. Formally, for learner i, the learning style is denoted by $C(i)$.

In order to effectively fuse the learning style (denotes the global semantics) and the behavior record (denotes the local semantics), the learning style is concatenated with the behavior record of each week in each registration (which is represented by Eq. 1) of the user. The concatenation will be input into a module of CNNs to derive the representations for predicting dropout behavior.

In Figure 2, we illustrate the process of the module on the registration of learner i on course j. The t-th row of the input matrix is the concatenation of the learning style ID of user i and the behavior record of learner i in the t-th week. From Figure 2, we know that the module of CNNs consists of one convolution layer, one pooling layer, and three fully connected layers. The convolution layer includes six feature maps, each of which is followed a pooling operation. Moreover, the kernel sizes of six feature maps are all set as 2×2, and the sizes of six max pooling operations are also set as 2×2. For the three fully connected layers, the number of units are set as 256, 128, and 32, respectively.

Given each registration of learner i on course j, the output of the CNNs is a matrix $h_{i,j} \in \mathbb{R}^{32 \times T}$, each column of which denotes the hidden representation of behavior in one week. Consequently, $h_{i,j}$ will be used for dropout prediction in the following subsection.

D. Dropout Prediction

Since the task of dropout prediction is a binary classification problem, a Sigmoid function is applied on top of the module of CNNs to derive the probability of dropout behavior. More specifically, given $h_{i,j}$ for registration of learner i on course j, the predicted probability of dropout in the $t \in \{1, \cdots, T\}$ is derived by the following equation.

$$\hat{y}_{i,j}^{(t)} = Sigmoid(W \cdot h_{i,j}^{(t)} + b) \tag{7}$$

where $h_{i,j}^{(t)}$ denotes the representation for the t-th week, and W and b denote the trained parameters.

To optimize the whole model, the cross-entropy loss function is used in our study. Given the training set of N registrations, and for each registration, there are T weeks of behavior records, then the loss function is defined as follows.

$$Loss = -\frac{1}{N} \sum_{(i,j)} \sum_{t=1}^{T} \left(y_{i,j}^{(t)} \cdot \log \hat{y}_{i,j}^{(t)} + (1 - y_{i,j}^{(t)}) \cdot \log(1 - \hat{y}_{i,j}^{(t)}) \right) \tag{8}$$

where $y_{i,j}^{(t)}$ denotes the ground-truth label for the t-th week of the registration of learner i on course j.

Through minimizing the loss function defined in Eq. 8, all parameters in the proposed model are optimized by the Adam optimizer in an end-to-end manner.

IV. Experiments

In this section, we conduct comprehensive experiments to evaluate the proposed methods from three aspects, including the overall performance comparison with selected baselines, ablation study, and the analysis of identified learning styles.

A. Dataset

In our experiments, the publicly available dataset "KDD Cup 2015" [1] is used to evaluate the proposed method. The original "KDD Cup 2015" includes 8,157,277 registrations (i.e., the learner-course pairs) of 79,186 students or learners on 39 courses. For each course, the time span of study process is five weeks. The statistics of the dataset is outlined in Table II. For each registration, the log file sequentially records various types of events made by the learner. There are seven types of learning behaviors in "KDD Cup 2015", and their descriptions are summarized in Table III.

Since predicting future dropout behavior requires a learner's historical learning information, learners with less than three

[1] https://www.biendata.xyz/competition/kddcup2015/data/

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

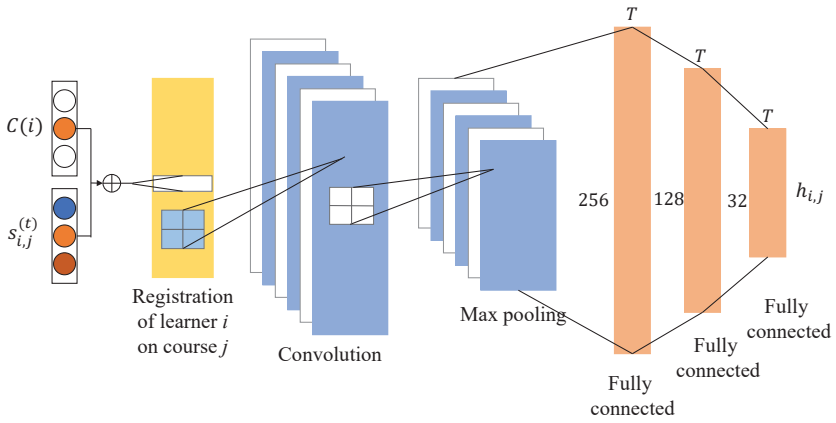


Fig. 2: The Framework of CNNs for Behavior Modeling.

registrations were removed in our experiments. After this filtering process, the final dataset consists of 37,461 registrations involving 9,063 learners and 39 courses. Based on the third definition shown in subsection II-A, there are four labels (i.e., each of the first four weeks will be assigned a label) in each registration, and the final dataset consists of 100,119 positive samples (i.e., dropouts) and 49,725 negative samples (i.e., non-dropouts). It means the ratio of positive samples to negative samples is about 7 to 3, and the imbalanced dataset will be a big challenge for optimizing the prediction models.

TABLE II: Statistics of KDD Cup 2015.

Item	Count
Courses	39
Students	79,186
Registration	8,157,277
Time Span	5 weeks

TABLE III: Event Types and Descriptions of KDD CUP 2015.

Event Type	Description
access	Access course-related material
discussion	Participate in course discussion
navigate	Access material other than courses
page close	Close page
problem	Complete post-lecture assignments
video	Watch course videos
wiki	View course-related wiki

B. Experimental Setup

To validate the effectiveness of our method, we select three traditional machine learning-based methods and two deep learning-based methods. Specifically, three traditional machine learning-based methods include FSPred [10], WSVM [11], and SPRF [23], which are implemented based on Logistic Regression, Support Vector Machine and Random Forest, respectively. As for deep learning-based methods, the selected baselines include NSSM [13] and LCNN [12], which are implemented based on LSTM and CNN, respectively.

Since dropout prediction is typically a binary classification task, we use the commonly used evaluation metrics for binary classification problem, including accuracy, precision, recall, and F1 score in our experiments.

In our experiments, the KDD CUP 2015 dataset is randomly divided into 70% for training and 30% for testing. In the process of partitioning dataset, it is guaranteed that the courses and learners occur in the testing set are also included in the training set, making sure the fluent completeness of the testing phase. In addition, we construct the procedure of the random dataset dividing 20 times, and the average performance of the 20 iterations is computed for each model for performance comparison.

C. Experimental Results and Analysis

1) Overall Performance Comparison: In order to evaluate our method, we select five representative models as baselines, and the results of overall performance comparison are presented in Table IV. Note that "Ours-OTS" denotes our model with OTS-based method for learning style detection, and "Ours-GAE" denotes our model with GAE-based method for learning style detection.

Results in Table IV clearly show that the proposed model outperforms all baselines in terms of all evaluation metrics. More specifically, all baselines achieve high recall and extremely low precision. Thereafter, the derived F1 score and accuracy are undesirable for all baselines. The main reason is that the imbalanced dataset leads to the common issue of high false positive rate for all baselines.

On the contrary, the proposed models (i.e., Ours-OTS and Ours-GAE) derive drastically enhanced precision, obtaining accordingly the better F1 score and accuracy on dropout prediction. There are two possible reasons for this improvement.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

TABLE IV: Results of Performance Comparison on KDD Cup 2015.

Methods	Accuracy	Precision	Recall	F1 Score
FSPred	0.7176	0.7165	0.9536	0.8183
WSVM	0.7133	0.7117	0.9582	0.8167
SPRF	0.7150	0.7305	0.9070	0.8093
NSSM	0.7077	0.7205	0.9746	0.8164
LCNN	0.7914	0.7941	0.9743	0.8517
Ours-OTS	**0.9558**	**0.9496**	0.9965	**0.9678**
Ours-GAE	0.9524	0.9392	**0.9978**	0.9655

On the one hand, by considering the learning styles of students, our model can effectively distinguish the dropout and non-dropout behavior based on the learning records in the studying process on MOOCs. On the other hand, the module of CNNs can derive more meaningful hidden representations by using global and local semantics in the learning behavior records. Moreover, Ours-OTS performs a little bit better than Ours-GAE, and the performance improvement is attributed to that semantics in the sequential context is fully employed by the OTS-based learning style detection.

2) Ablation Study: In this section, we test the contribution of learning style detection and the module of CNNs to the improvement of the proposed model. Specifically, we built three variants of our model with OTS-based feature extraction, and the results are presented in Table V. In Table V, "OTS w/o L.S." represents the variant where the learning style detection is removed from our model, while "OTS w/o CNNs" denotes the variant in which the module of CNNs is removed form our model. "OTS w/o Both" represents the variant in which both learning style detection and the module of CNNs are removed in our model, almost reducing to a logistic regression model. Note that our model with GAE-based feature extraction derives the similar results of case study, which are not shown due to the limit of space.

TABLE V: Results of Ablation Study.

Model	Accuracy	Precision	Recall	F1 Score
OTS w/o. Both	0.7073	0.6987	0.9576	0.8079
OTS w/o. L.S.	0.7149	0.7255	0.9644	0.8281
OTS w/o. CNNs	0.7913	0.7887	0.9547	0.8638
OTS	**0.9558**	**0.9496**	0.**9965**	**0.9678**

The results in Table V suggest that both components (i.e., learning style detection and the module of CNNs) are essential for improving the performance of the proposed model, and the variant "OTS w/o. Both" derives the worse performance. Without the learning style detection, the F1 score of the proposed model drastically drops from 0.9678 to 0.8281, indicating that the learning style detection plays a significant role for the proposed model. Comparing with the module of learning style detection, the module of CNNs contributes less for the performance enhancement of the proposed model. But the results show that the module of CNNs is also important for our model. Therefore, for the proposed model, both learning style detection and the module of CNNs are required to derive desirable results of dropout prediction.

3) Analysis of Learning Style Detection: Since learning style detection is crucial for our model, in this section, we visualize the obtained learning styles in our experiments. More specifically, based on the elbow method [29], eight clusters of learners are obtained, each of which is viewed as a group with the common learning styles. For each cluster, we plot the average frequency of different event types in each week. The eight learning styles obtained by OTS-based method are shown in Figure 3. Note that in each subfigure of Figure 3, the horizontal axis represents five weeks of the studying process, and the vertical axis illustrates the average frequencies of seven learning behaviors for learners sharing each learning style.

From Figure 3, we have two observations. On the one hand, learners exhibit indeed different learning styles, and each learning style can be expressed by different patterns of learning behaviors. On the other hand, even for the same learning style, the patterns of learning behaviors are distinct in different weeks of the studying process. Therefore, the DTW-based K-means clustering is also important for our model to derive desirable results on the task of dropout prediction.

V. Conclusions

This paper has proposed a novel deep learning-based model for dropout prediction, in which students' learning styles are introduced for behavior modeling. Specifically, two feature extraction methods, i.e., OTS-based method and GAE-based method, are employed to obtain the initial features. Then, a dynamic time warping-based K-means clustering is utilized to identify distinctive learning styles on MOOCs. Finally, in order to capture the semantics in global and local contexts, a module of CNNs is used to derive more meaningful representations, on which the dropout prediction is conducted. The results of comprehensive experiments demonstrate that the proposed method performs better than representative baselines. Moreover, the results of ablation study indicate the contributions of the two components in our method, i.e., both learning style detection and the module of CNNs are required for the proposed method. In addition, by visualizing different patterns of learning behaviors, we illustrate further the importance of the learning style detection for dropout prediction.

Acknowledgement

This work was partially supported by the National Natural Science Foundation of China (62472192), the Fundamental Research Funds for Central Universities (CCNU24JC032) and Natural Science Foundation of Hubei Province of China (2022CFB289).

References

[1] J. Daniel, "Making sense of moocs: Musings in a maze of myth, paradox and possibility," *Journal of interactive Media in education*, vol. 2012, no. 3, 2012.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.

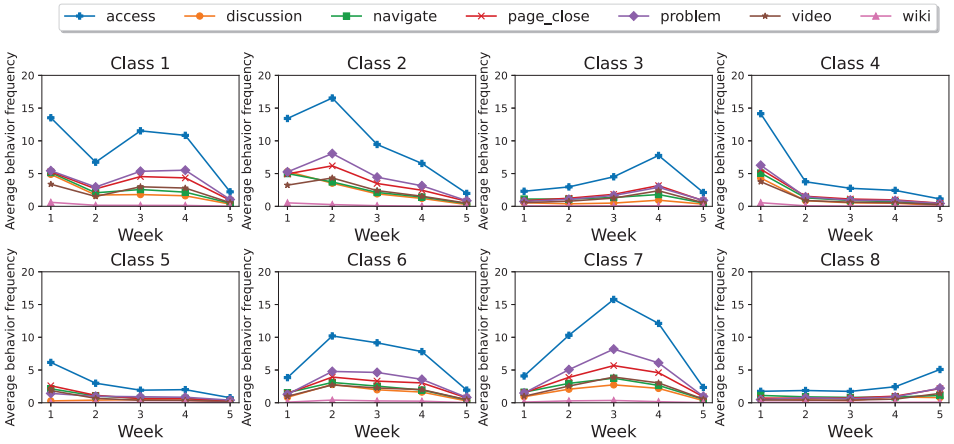


Fig. 3: Visualization of learning styles based on OTS-based method.

[2] H. B. Shapiro, C. H. Lee, N. E. W. Roth, K. Li, M. Çetinkaya-Rundel, and D. A. Canelas, "Understanding the massive open online course (mooc) student experience: An examination of attitudes, motivations, and barriers," *Computers & Education*, vol. 110, pp. 35–50, 2017.

[3] Y. Li, "Moocs in higher education: opportunities and challenges," in *2019 5th international conference on humanities and social science research (ICHSSR 2019)*. Atlantis Press, 2019, pp. 48–55.

[4] K. Ronkowitz and L. C. Ronkowitz, "Moocs: Evolution and revolution," in *Macro-level learning through massive open online courses (MOOCs): Strategies and predictions for the future*. IGI Global, 2015, pp. 183–211.

[5] J. Goopio and C. Cheung, "The mooc dropout phenomenon and retention strategies," *Journal of Teaching in Travel & Tourism*, vol. 21, no. 2, pp. 177–197, 2021.

[6] H. Liu, X. Chen, and F. Zhao, "Learning behavior feature fused deep learning network model for mooc dropout prediction," *Education and Information Technologies*, vol. 29, no. 3, pp. 3257–3278, 2024.

[7] T. Pan, G. Feng, X. Liu, and W. Wu, "Using feature interaction for mining learners' hidden information in mooc dropout prediction," in *International Conference on Intelligent Tutoring Systems*. Springer, 2023, pp. 507–517.

[8] C. Jin, "Mooc student dropout prediction model based on learning behavior features and parameter optimization," *Interactive Learning Environments*, vol. 31, no. 2, pp. 714–732, 2023.

[9] F. Dalipi, A. S. Imran, and Z. Kastrati, "Mooc dropout prediction using machine learning techniques: Review and research challenges," in *2018 IEEE global engineering education conference (EDUCON)*. IEEE, 2018, pp. 1007–1014.

[10] L. Qiu, Y. Liu, and Y. Liu, "An integrated framework with feature selection for dropout prediction in massive open online courses," *IEEE Access*, vol. 6, pp. 71 474–71 484, 2018.

[11] M. Kloft, F. Stiehler, Z. Zheng, and N. Pinkwart, "Predicting mooc dropout over weeks using machine learning methods," in *Proceedings of the EMNLP 2014 workshop on analysis of large scale social interaction in MOOCs*, 2014, pp. 60–65.

[12] Y. Wen, Y. Tian, B. Wen, Q. Zhou, G. Cai, and S. Liu, "Consideration of the local correlation of learning behaviors to predict dropouts from moocs," *Tsinghua Science and Technology*, vol. 25, no. 3, pp. 336–347, 2019.

[13] F. Wang and L. Chen, "A nonlinear state space model for identifying at-risk students in open online courses." *International Educational Data Mining Society*, 2016.

[14] K. Talebi, Z. Torabi, and N. Daneshpour, "Ensemble models based on cnn and lstm for dropout prediction in mooc," *Expert Systems with Applications*, vol. 235, p. 121187, 2024.

[15] J.-L. Hung, M. C. Wang, S. Wang, M. Abdelrasoul, Y. Li, and W. He, "Identifying at-risk students for early interventions—a time-series clustering approach," *IEEE Transactions on Emerging Topics in Computing*, vol. 5, no. 1, pp. 45–55, 2015.

[16] W. He, "Examining students' online interaction in a live video streaming environment using data mining and text mining," *Computers in Human Behavior*, vol. 29, no. 1, pp. 90–102, 2013.

[17] D. Yang, T. Sinha, D. Adamson, and C. P. Rosé, "Turn on, tune in, drop out: Anticipating student dropouts in massive open online courses," in *Proceedings of the 2013 NIPS Data-driven education workshop*, vol. 11. Lake Tahoe, NV, 2013, p. 14.

[18] J. Gardner, C. Brooks, and R. Baker, "Evaluating the fairness of predictive student models through slicing analysis," in *Proceedings of the 9th international conference on learning analytics & knowledge*, 2019, pp. 225–234.

[19] J. Gardner, Y. Yang, R. S. Baker, and C. Brooks, "Modeling and experimental design for mooc dropout prediction: A replication perspective." *International Educational Data Mining Society*, 2019.

[20] W. Xing, X. Chen, J. Stein, and M. Marcinkowski, "Temporal predication of dropouts in moocs: Reaching the low hanging fruit through stacking generalization," *Computers in human behavior*, vol. 58, pp. 119–129, 2016.

[21] J. Whitehill, K. Mohan, D. Seaton, Y. Rosen, and D. Tingley, "Mooc dropout prediction: How to measure accuracy?" in *Proceedings of the fourth (2017) acm conference on learning@ scale*, 2017, pp. 161–164.

[22] X. Zhang, X. Wang, J. Zhao, B. Zhang, and F. Zhang, "Ic-btcn: A deep learning model for dropout prediction of moocs students," *IEEE Transactions on Education*, 2024.

[23] S. Dass, K. Gary, and J. Cunningham, "Predicting student dropout in self-paced mooc course using random forest model," *Information*, vol. 12, no. 11, p. 476, 2021.

[24] B. A. Muhammad, C. Qi, Z. Wu, and H. K. Ahmad, "Grl-ls: A learning style detection in online education using graph representation learning," *Expert Systems with Applications*, vol. 201, p. 117138, 2022.

[25] H. Zhang, T. Huang, S. Liu, H. Yin, J. Li, H. Yang, and Y. Xia, "A learning style classification approach based on deep belief network for large-scale online education," *Journal of cloud computing*, vol. 9, pp. 1–17, 2020.

[26] N. Hidayat, R. Wardoyo, A. Sn, and H. D. Surjono, "Enhanced performance of the automatic learning style detection model using a combination of modified k-means algorithm and naive bayesian," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 3, pp. 638–648, 2020.

[27] T. N. Kipf and M. Welling, "Variational graph auto-encoders," in *Proceedings of NeurIPS Workshop on Bayesian Deep Learning (NeurIPS BDL)*, 2016.

[28] F. Petitjean, A. Ketterlin, and P. Gançarski, "A global averaging method for dynamic time warping, with applications to clustering," *Pattern recognition*, vol. 44, no. 3, pp. 678–693, 2011.

[29] M. Cui *et al.*, "Introduction to the k-means clustering algorithm based on the elbow method," *Accounting, Auditing and Finance*, vol. 1, no. 1, pp. 5–8, 2020.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on April 19,2026 at 09:55:22 UTC from IEEE Xplore. Restrictions apply.